CS383 · SOFTWARE ENGINEERING

Practice Question Bank

Practice on the midterm's question types · 8 questions

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01 Before you start

These questions are my own, in the same types as the four midterms, with new texts and numbers. The ethics statements are taken from the text of the IEEE Code itself. The solutions are in a separate file.

- Q1: ethics. Q2 and Q3: choosing the model. Q4: sprint calculations.
- Q5 to Q7: requirements. Q8: short questions from all the lectures.

02 Questions

Q1 ETHICS

According to the 8 principles of the IEEE Code of Ethics (PUBLIC, CLIENT AND EMPLOYER, PRODUCT, JUDGMENT, MANAGEMENT, PROFESSION, COLLEAGUES, SELF), identify the type of ethics reflected in the following descriptions:

SEEPP ethic example	Ethics Principle
1. Keep private any confidential information gained in their professional work.	
2. Ensure adequate testing, debugging, and review of software and related documents on which they work.	
3. Not engage in deceptive financial practices such as bribery, double billing, or other improper financial practices.	
4. Offer fair and just remuneration.	
5. Promote public knowledge of software engineering.	
6. Credit fully the work of others and refrain from taking undue credit.	
7. Further their knowledge of developments in the analysis, specification, design, development, maintenance and testing of software.	
8. Disclose to appropriate persons or authorities any actual or potential danger to the user, the public, or the environment.	

Q2 PLAN-DRIVEN OR AGILE

For each concern, determine whether the Plan Driven or Agile method is the most suitable choice:

Concern	Plan Driven	Agile
1- Detailed documentation is required before coding starts		
2- The customer is available to the team every day		
3- A fixed-price contract with a fixed scope		
4- Working software is delivered every two weeks		
5- Requirements are expected to change frequently		
6- Formal sign-off at the end of each phase		

Q3 DEVELOPMENT MODEL

For each project, choose one development model (Agile, Plan-Driven, Iterative, or Component-Based) and justify your choice.

Project 1 — Hospital Billing Integration. Requirements are moderately stable. The system must be assembled from an existing certified payment gateway, an off-the-shelf invoicing package and the insurance company's SDK. 12 engineers.

Project 2 — Air-Traffic Control Module. Safety-critical. Stakeholders define the requirements up front and review at stage gates only. Requirements are fixed by aviation regulation. 40 engineers in three sites; every decision must be documented and signed off.

Project 3 — Food-Delivery Startup MVP. The founder sits with the team and gives daily feedback. Features change every week. 5 co-located developers. A first release is needed in one month.

Project 4 — University Registration Upgrade. The registrar's office is available for a review every month. The core requirements are stable, but some changes are expected. 15 co-located developers. The product needs regular iterations to refine it.

Q4 SPRINT CALCULATIONS

Team of 5 people; workday = 8h. Last 5 sprints completed story points: 30, 34, 28, 32, 36.

1. Compute the sprint velocity.
2. Compute the focus factor.
3. For a 2-week sprint (weekends off), compute the effective team capacity in hours.
4. If the next sprint backlog is 33 story points, is the plan feasible? What if it is 50 story points?

Q5 USER / SYSTEM · FR / NFR

For each statement below, classify (i) User vs. System requirement and (ii) Functional (FR) vs. Non-functional (NFR).

Requirement	User vs. System	FR vs. NFR
a) "The app should be secure."		
b) "The system shall lock the account after 5 failed login attempts within 10 minutes."		
c) "Students can view their grades."		
d) "The search page shall return results within 2 seconds for 95% of requests."		

Q6

PRODUCT / ORGANIZATION / EXTERNAL

For each statement, indicate the requirement type: Product, Organization or External.

Statement	Type
1. The system shall be available 99.5% of the time during working hours.	
2. All source code must follow the company's coding standard and be stored on the company's Git server.	
3. The system must comply with the Personal Data Protection Law.	
4. The mobile app shall use no more than 150 MB of memory.	
5. The project must be delivered using the organization's Scrum process with two-week sprints.	
6. The payment module must conform to the PCI DSS standard.	

Q7 FR AND NFR

Extract 4 functional requirements and 3 non-functional requirements from the following description.

A university library needs an online system. Students can search for books by title or author and reserve an available book. Librarians can add new books and record returned books. The system must be available 24 hours a day, and search results must appear within 3 seconds. All passwords must be stored encrypted.

Q8 SHORT QUESTIONS

1. What are the 4 key software engineering activities, in a logical order?
2. Give one example of a generic product and one example of a customized product.
3. State two weaknesses of the waterfall model.
4. What are the four factors that product quality depends on?
5. What are the four factors a project manager considers in people management?
6. What are the inputs and the outputs of sprint planning?
7. True or False, with a reason: "Software development costs more than software maintenance."
8. True or False, with a reason: "The initial estimation of effort is always accurate."
9. In XP, what is the practice of continuously improving the code called?
10. State one benefit and one drawback of Software as a Service (SaaS).